

$$h_{fb} = \frac{q''_{fb}}{\left[(T_{w,CHF} - T_{sat}) + \left| \left(\frac{q''_{fb} - q''_{CHF}}{q''_{min} - q''_{CHF}} \right)^{0.8} \right| \{ (T_{w,min} - T_{sat}) - (T_{w,CHF} - T_{sat}) \} \right]} \quad (24)$$

It should be noted that Eq. (24) is structurally identical to the equation used in the baseline study [14]. However, the TB regime is modified in the present work by virtue of the updated correlations for CHF and MHF, which directly influence the input parameters used in this formulation.

2.8. Film boiling (FB)

Film boiling (FB) is recognized as the highest temperature regime in pool boiling, wherein the high surface temperature causes bubble nucleation to completely subside and be replaced by the formation of a continuous vapor blanket. A detailed description of this regime and its characteristics has been provided in the authors' recently published work [16]. In the same study, the following baseline HTC correlation was derived based on available data for cryogenic fluids:

$$h_{fb} = (0.148 + 0.052 \sin\theta) \frac{k_g}{L_b} \left[Ra_g \left(\frac{h_{fg} + 0.46 c_{p,g} (T_{w,fb} - T_f)}{c_{p,g} (T_{w,fb} - T_f)} \right) \right]^{0.33} + h_r \quad (25)$$

where h_r denotes the radiation HTC, expressed by the following relation:

$$h_r = \frac{(1.7 - 0.55 \sin\theta) \sigma_b (T_{w,fb}^4 - T_f^4)}{(T_{w,fb} - T_f)} \quad (26)$$

In Eq. (25), h_{fb} represents the overall film boiling (FB) HTC [$W \cdot m^{-2} \cdot K^{-1}$], accounting for both conduction through the vapor film and thermal radiation effects. The angle θ refers to the orientation angle of the heated surface measured from the horizontal upward position [$^\circ$], k_g is the thermal conductivity of saturated vapor [$W \cdot m^{-1} \cdot K^{-1}$], $T_{w,fb}$ is the temperature of the heating surface in the film boiling regime [K], and σ_b is the Stefan-Boltzmann constant, $5.67 \times 10^{-8} W \cdot m^{-2} \cdot K^{-4}$. The term L_b , known as the Laplace constant or bubble length [m], is a characteristic length scale that governs vapor film stability and curvature effects, and is calculated using the following relation:

$$L_b = \sqrt{\frac{\sigma}{g(\rho_f - \rho_g)}} \quad (27)$$

Moreover, Ra_g is the Rayleigh number at saturated vapor conditions, which can be determined as:

$$Ra_g = \frac{L_b^3 \rho_g (\rho_f - \rho_g) g}{\mu_g^2} \left(\frac{\mu_g c_{p,g}}{k_g} \right) \quad (28)$$

Although the above baseline FB correlation achieved an MAE of 12.94 %, it does not account for effects beyond surface orientation and radiation. With the incorporation of recently acquired experimental data and a comprehensive reassessment of the universal database, the FB correlation is further refined to include these additional influencing parameters. This enhancement is aimed at improving predictive accuracy across a broader range of conditions, particularly those relevant to cryogenic thermal system design. The various factors affecting the FB regime will be discussed individually in the following sections, along with the formulation of corresponding multipliers or correction functions to be applied to the baseline correlation.

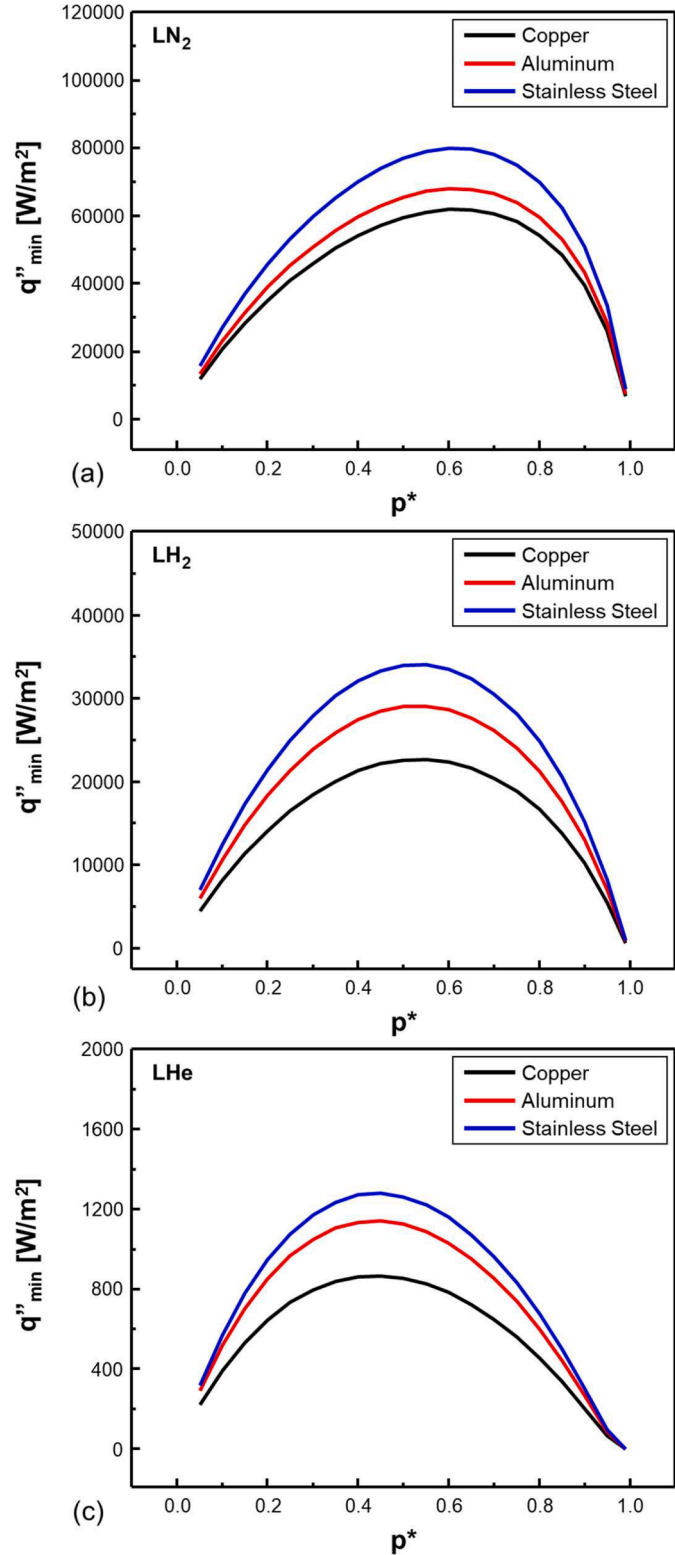


Fig. 13. Variation of q''_{min} with reduced pressure for different surface materials predicted according to newly modified correlation for (a) LN_2 , (b) LH_2 , and (c) LHe .